AMANDA ROBINSON

Software Engineer

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SUMMARY

Junior Computer Science student at Worcester Polytechnic Institute. Pursuing internships in software engineering, product management, data analysis, or UX design. Experienced in full-stack development, relational database design, and healthcare data systems, with a track record of building tools that turn raw data into usable products.

EDUCATION

Worcester Polytechnic Institute *Worcester, MA*

B.S. Computer Science, Minor in Economics Expected December 2027

Dean's List: Fall 2024, Fall 2025

Relevant Coursework: Software Engineering, Object Oriented Analysis and Design, Database Systems, Machine Learning, Human-Computer Interaction, Data Science, Algorithms, Operating Systems, Systems Programming, Machine Organization/Assembly Language, Cryptography

EXPERIENCE

South Florida Proton Therapy Institute - Software Intern *May 2026 - Aug 2026*

Delray Beach, FL

- Engineered an application to automate MRI/CT scan downloads, supporting the testing of a program built to improve patient treatment safety
- Isolated relevant clinical data in the Varian medical database while maintaining strict HIPAA compliance
- Built dynamic heatmap visualizations from geospatial data to help the company identify priority regions for outreach

  Technologies: *SQL Server, SQL, DICOM, Java, Python, TypeScript, React, Next.js, Leaflet, GeoJSON, Google Places API*

Threekit - Intern *Jul 2022 - Aug 2022*

Chicago, IL (Remote)

- Prototyped a path-tracing rendering technique for e-commerce product catalogs under the mentorship of CTO Ben Houston

  Technologies: *HTML, JavaScript, Three.js*

PROJECTS

Sentiment to Stars *WPI CS4342 - Machine Learning*

- Built an NLP model that predicts 1–5 star ratings from review text alone using 500K+ Amazon reviews, reaching 82.5% accuracy
- Replaced an RNN + linear regression baseline by fine-tuning DistilBERT via transfer learning; built a React frontend for live demos

  Technologies: *Python, PyTorch, Hugging Face Transformers, DistilBERT, TypeScript, React*

Yelp Business Search Application *WPI CS3431 - Database Systems*

- Built a search application spanning 150K+ Yelp businesses, enabling discovery by location, category, and custom attributes alongside user reviews
- Designed a normalized relational schema from an ER model and deployed it to PostgreSQL via SQL DDL
- Implemented complex queries, views, and triggers to enforce referential integrity and business rules
- Developed a dual-language client, pairing a Java backend with a Python/PySide6 GUI, to connect user actions to live database queries

  Technologies: *PostgreSQL, SQL, Java, Python, PySide6*

Shopping Comparison Service *WPI CS3733 - Software Engineering*

- Built a full-stack grocery comparison web app for receipt management, shopping lists, and best-price computation from historical purchase data
- Designed and refactored a MySQL relational schema and developed the complex queries powering core app logic
- Implemented business logic and infrastructure as code on AWS Lambda; designed the UI in Figma and coordinated team execution

  Technologies: *Next.js, React, TypeScript, AWS Amplify/Cognito, AWS CDK, MySQL*

NextFlex FlexFactor Competition *Oct 2023 - Dec 2023*

- Built MEMO, a wrist-worn time-management device using MicroPython on a Raspberry Pi Pico with GPIO; placed 3rd in Santa Clara County

intervle.fun *Summer 2023*

- Developed a responsive, Wordle-inspired web game with a lexicographic twist using HTML, CSS (Bootstrap), and JavaScript

SKILLS

Languages: Java, Python, SQL, TypeScript, HTML/CSS, C++, C, Racket

Frameworks/Libraries: React.js, Next.js, Node.js/Express, PyTorch, Hugging Face Transformers, PySide6, Bootstrap

Tools: Git/GitHub, VS Code, Claude Code, Linux, Raspberry Pi

Databases: MySQL, PostgreSQL, SQL Server Management Studio

Other: Machine Learning, NLP, AWS, Figma, MicroPython, UX Principles

LEADERSHIP & ACTIVITIES

- Alpha Gamma Delta Sorority, Interim VP of New Member Experience - Planned and executed Bid Day and pledge-in events for incoming members
- Masque Theater, VP of Props & Costumes - Managed the props & costumes closet for 3 years; created documentation for all department positions
- Women's Club Rugby, PR Officer - Designed Instagram content to recruit prospective team members
- Girls Who Code (Leigh High School), Co-President - National Merit Finalist

INTERESTS & HOBBIES

Digital Photography, Theater, Logic Puzzles, Board Games, Baking, Origami, Rugby